A Log-Phase Mapping of Physical Constants Using the First Zero of the Riemann Zeta Function

Jacob Powell

2026

Abstract

A numerical coincidence between cosmological scale ratios and the first non-trivial zero of the Riemann zeta function suggests a logarithmic mapping between physical constants and the imaginary part of the zeta zero.

Specifically,

$$\ln(R/\ell_P) \approx 10\gamma_1$$

where R represents a cosmological length scale, ℓ_P is the Planck length, and γ_1 is the first non-trivial zero of the Riemann zeta function.

Motivated by this observation, a logarithmic phase mapping of physical constants relative to Planck-scale normalization is defined:

$$\phi(Q) = \frac{\ln(Q/Q_P)}{10\gamma_1} \mod 1$$

Using a dataset of 39 physical constants including particle masses, dimensionless couplings, thermal scales, and gravitational couplings, clustering appears near the fractional phases 0, 1/3, 1/2, and 2/3.

Monte Carlo tests under a simplified random-phase model indicate that clustering of comparable strength occurs rarely (10^{-5} frequency). While this does not constitute evidence of a fundamental law, the mapping provides an exploratory phenomenological framework for studying logarithmic hierarchies in physics.

1 Introduction

Large hierarchy gaps appear throughout modern physics. Ratios between cosmological length scales and the Planck scale reach values near

$$\ln(R/\ell_P) \approx 141.46$$

where R represents the approximate radius of the observable universe ($R \sim 4.4 \times 10^{26}\,\mathrm{m}$) and ℓ_P is the Planck length.

An intriguing observation is that this value is extremely close to

$$10\gamma_1$$

where

$$\gamma_1 \approx 14.134725$$

is the first non-trivial zero of the Riemann zeta function.

Connections between spectral structures and the zeros of the zeta function have long been discussed in the context of the Hilbert–Pólya conjecture, which proposes that the non-trivial

zeros correspond to eigenvalues of a self-adjoint operator. While the present work does not attempt to derive such an operator, the appearance of the first zeta zero in a logarithmic phase mapping motivates exploratory investigation.

2 Logarithmic Phase Mapping

Define the phase coordinate

$$\phi(Q) = \frac{\ln(Q/Q_P)}{P} \mod 1$$

where

$$P = 10\gamma_1 \approx 141.3472514173$$

and Q_P represents the appropriate Planck-scale normalization for the physical quantity Q (for example Planck mass, Planck temperature, or Planck coupling).

This transformation maps quantities spanning many orders of magnitude into a circular coordinate on the interval $[0, 1)$ in logarithmic space.

3 Data

The dataset used in this analysis contains 39 widely cited physical constants drawn from standard sources including CODATA, the Particle Data Group, and cosmological measurements.

Constants are grouped into four sectors:

- Dimensionless couplings
- Particle masses
- Thermal / neutrino scales
- Gravitational couplings

The full dataset, analysis scripts, and reproduction code are available at
https://github.com/Jacob3175/ZetaPhase

4 Observed Phase Structure

Applying the mapping $\phi(Q)$ reveals clustering near simple fractional phase values.

Sector	Count	Mean ϕ	Std dev	Mean dist to band	Max dist to band
Dimensionless	16	0.0273	0.0358	0.0273	0.1502
Massive particles	16	0.3104	0.0294	0.0314	0.0614
Thermal / Neutrino	5	0.5024	0.0172	0.0144	0.0213
Gravitational	2	0.6759	0.0752	0.0532	0.0624

Table 1: Sector statistics from the 39-constant dataset.

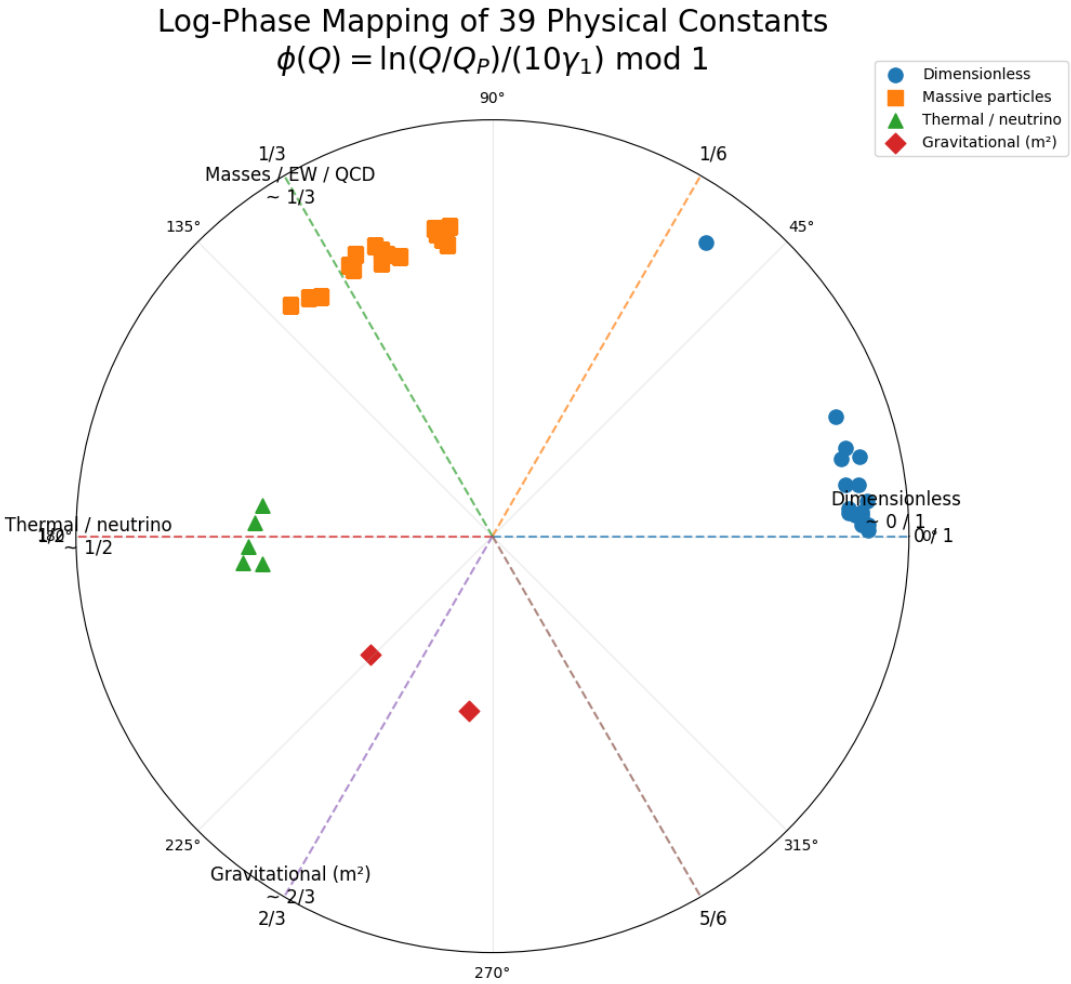

Figure 1: Log-phase mapping $\phi(Q)$ of the constants dataset. Blue points represent constants while dashed red lines mark the fractional phase centers 0, 1/3, 1/2, and 2/3.

5 Statistical Tests

Monte Carlo simulations were performed by generating 100,000 randomized phase distributions while preserving sector counts.

Clustering comparable to the observed dataset occurred rarely (10^{-5} frequency) under this simplified random-phase model.

Sensitivity scans over scaling constants C in

$$P = C\gamma_1$$

show that clustering metrics are minimized near $C \approx 10$.

6 Blind Prediction Test

To test whether the observed clustering reflects genuine structure rather than post-hoc pattern recognition, a blind prediction test can be performed.

The dataset is divided into training and holdout groups. Phase band centers are determined using only the training constants. The holdout constants are then evaluated relative to these predicted bands.

If holdout constants remain near the predicted phase bands, this would provide stronger evidence that the clustering is not solely the result of dataset selection.

7 Limitations

The dataset contains a limited number of constants and correlations between constants are not fully accounted for. Sector boundaries were defined using conventional physical classifications rather than blind clustering.

Consequently the results should be interpreted as exploratory rather than definitive.

8 Future Work

Future research directions include:

- Expansion to larger constant datasets
- Blind prediction testing
- Investigation of higher zeta zeros
- Exploration of theoretical connections between number-theoretic spectra and physical scale hierarchies

9 Conclusion

A logarithmic phase mapping based on the first non-trivial zero of the Riemann zeta function produces structured clustering among several classes of physical constants.

Although the mapping does not currently derive from known physical theory, the persistence of the pattern and the numerical proximity between $10\gamma_1$ and the cosmological hierarchy $\ln(R/\ell_P)$ suggest that the approach may offer a useful exploratory framework for investigating scale hierarchies in physics.

References

[1] B. Riemann, *On the Number of Primes Less Than a Given Magnitude*, 1859.

[2] Particle Data Group, *Review of Particle Physics*, Prog. Theor. Exp. Phys.

[3] CODATA, *Recommended Values of the Fundamental Physical Constants.*